



MA 575: Complex Analysis

Course Code	MA 575
Course Name	Complex Analysis
L-T-P-C	3-0-1-4
Prerequisites	None
Intended for	M.Sc./M.Tech./PhD/B.Tech
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Module 1:** Complex numbers and the point at infinity, Polar representation, logarithmic and trigonometric functions, Analytic functions, Cauchy-Riemann conditions, Power series, Harmonic conjugates, Mobius transformations, Mappings by elementary functions, Liouville's theorem and its applications, Conformal mappings. (14 Hours)
- **Module 2:** Index of a closed curve, Cauchy's theorem, Cauchy integral formula, Power series representation of analytic functions, Open mapping theorem, Goursat's theorem. Uniform convergence of sequences and series. Taylor and Laurent series. (12 Hours)
- **Module 3:** Isolated singularities and residues, Residue theorem and its application to evaluation of real integrals. Zeroes and poles, Maximum Modulus Principle and Schwarz's lemma, Meromorphic functions, Argument Principle, Rouché's theorem, (16 Hours)

Laboratory Modules

1. NA

Text Books

1. J.B. Conway, **Functions of One Complex Variable**, 2nd Edition, Narosa, New Delhi, 1978.
2. T.W. Gamelin, **Complex Analysis**, Springer International Edition, 2001.
3. J.W. Brown and R.V. Churchill, **Complex variables and applications**, 9th edition, McGraw Hill Higher Education.

References

1. R. Remmert, **Theory of Complex Functions**, Springer Verlag, 1991.
2. A.R. Shastri, **An Introduction to Complex Analysis**, Macmillan India, New Delhi, 1999.
3. E.M. Stein and R. Shakarchi, **Complex analysis**, Princeton lecture series in analysis.
4. M.Thamban Nair, **Complex analysis online notes**, <https://home.iitm.ac.in/mtnair/ca.pdf>